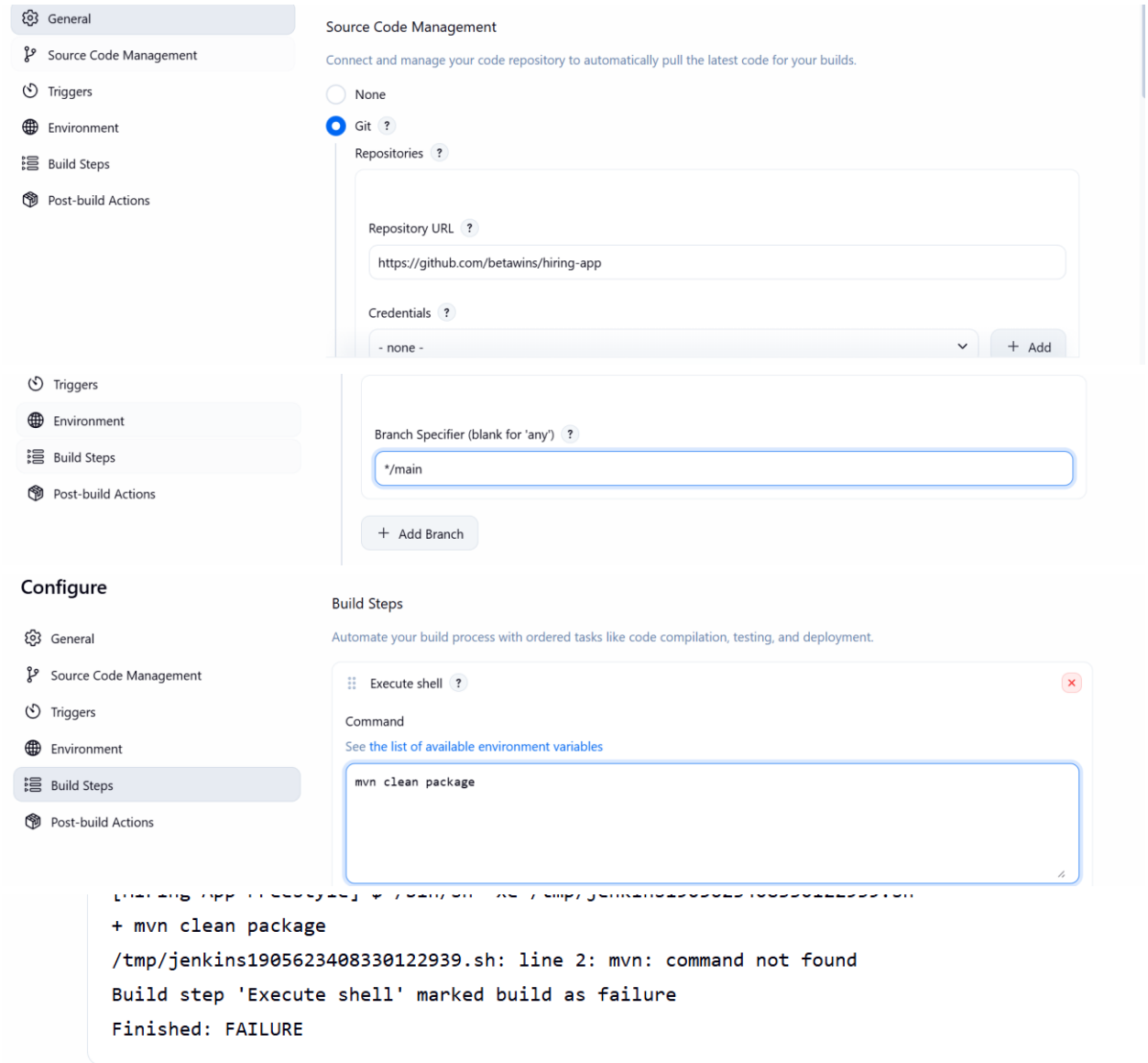
1) Setup Jenkins CICD pipeline using freestyle job using Docker containers using below code.

Source Code: https://github.com/betawins/hiring-app.git

Stages:

- **Git Clone**
- **SonarQube Integration**
- **Maven Compilation**
- **Nexus Artifactory**
- **Slack Notification**
- **Deploy On tomcat**



Failure because of no maven installed

```
Complete!
[root@ip-10-0-14-116 ~]# mvn --version
Apache Maven 3.8.4 (Red Hat 3.8.4-3.amzn2023.0.5)
Maven home: /usr/share/maven
```

Jenkins / Hiring-App-Freestyle / #2

```
filtering/3.1.1/maven-filtering-3.1.1.jar (51 kB at 156 kB/s)
Downloaded from central: https://repo.maven.apache.org/maven2/org/codehaus/plexus/plexus-utils/3.3.0/plexus-utils-3.3.0.jar (263 kB at 800 kB/s)
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/shared/maven-mapping/3.0.0/maven-mapping-3.0.0.jar (11 kB at 31 kB/s)
[INFO] Packaging webapp
[INFO] Assembling webapp [hiring] in [/var/lib/jenkins/workspace/Hiring-App-Freestyle/target/hiring]
[INFO] Processing war project
[INFO] Copying webapp resources [/var/lib/jenkins/workspace/Hiring-App-Freestyle/src/main/webapp]
[INFO] Building war: /var/lib/jenkins/workspace/Hiring-App-Freestyle/target/hiring.war
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  16.969 s
[INFO] Finished at: 2026-10-05T18:37:16Z
[INFO] ------------------------------------------------------------------------
Finished: SUCCESS
```

```
[root@ip-10-0-14-116 ~]# docker run -d --name nexus -p 8081:8081 -v nexus-data:/nexus-data sonatype/nexus
Unable to find image 'sonatype/nexus:latest' locally
latest: Pulling from sonatype/nexus
972401f623c8: Pull complete
fb5c53e59f6e: Pull complete
5d76986c3a82: Pull complete
341c1b34e3ea: Pull complete
4f4fb700ef54: Pull complete
Digest: sha256:6bc88b51d4d71d8959f03b6a3caba71f94c4aa50e709bf41f2bc6dc3a5d9f01c
Status: Downloaded newer image for sonatype/nexus:latest
73f6b9393b4d4b12da565c1bd0ab0ac6a49ac4804cdde1cfb9eed2597ab28492
```

Jenkins → SonarQube (9000) → Maven → Nexus (8081) → Tomcat

You don't need to install Docker again. We only need to start the Nexus container next.

So our Jenkins-2 server now has:

Jenkins :8080 → SonarQube :9000 → Nexus :8081

```
[root@ip-10-0-14-116 ~]# docker logs nexus 2>&1 | grep -i "started"
2026-10-05 18:40:08,943+0000 INFO  [jetty-main-1]  org.sonatype.nexus.apachehttpclient.Hc4ProviderImpl - Started (connectionPoolMaxSize 200,
 connectionPoolSize 20, connectionPoolIdleTime 30000 ms, connectionPoolTimeout 30000 ms, keepAliveMaxDuration 30000 ms, validateAfterInactiv
ityTime 2000 ms)
2026-10-05 18:40:09,108+0000 INFO  [jetty-main-1]  org.sonatype.timeline.internal.DefaultTimeline - Started Timeline...
2026-10-05 18:40:10,454+0000 INFO  [jetty-main-1]  org.sonatype.nexus.NxApplication - Started Nexus Repository Manager OSS 2.15.2-03
2026-10-05 18:40:12,384+0000 INFO  [jetty-main-1]  org.eclipse.jetty.server.handler.ContextHandler - Started o.e.j.w.WebAppContext@28b710b1{
Nexus Repository Manager 2.15.2-03,/nexus,file:///opt/sonatype/nexus/nexus/,AVAILABLE}{/opt/sonatype/nexus/nexus}
2026-10-05 18:40:12,440+0000 INFO  [jetty-main-1]  org.eclipse.jetty.server.AbstractConnector - Started ServerConnector@45f45fa1{HTTP/1.1, (
http/1.1)}{0.0.0.0:8081}
2026-10-05 18:40:12,441+0000 INFO  [jetty-main-1]  org.eclipse.jetty.server.Server - Started @14022ms
2026-10-05 18:40:12,442+0000 INFO  [main] *SYSTEM org.sonatype.nexus.bootstrap.jetty.JettyServer - Started
```

Download progress

Preparation

- Checking internet connectivity
- Checking update center connectivity
- Success

Nexus Artifact Uploader ✓ Success

Loading plugin extensions ✓ Success

Nexus artifact uploader

Nexus Details

Nexus Version

NEXUS2

Protocol

HTTP

Nexus URL ?

100.58.236.247:8081/nexus

GroupId

com.hiring

Version

1.0

Repository ?

releases

ArtifactId

hiring

Type ?

war

Classifier ?

File ?

target/hiring.war

```
Type: war
Version: 1.0
File: hiring.war
Repository:releases
Uploading: http://localhost:8081/nexus/content/repositories/releases/com/hiring/hiring/1.0/hiring-1.0.war
100 % completed (1.7 kB / 1.7 kB).
Uploaded: http://localhost:8081/nexus/content/repositories/releases/com/hiring/hiring/1.0/hiring-1.0.war (1.7
kB at 9.8 kB/s)
Uploading artifact hiring.war completed.
Finished: SUCCESS
```

i changed nexus url to localhost:8081

The build is success now

Your Jenkins pipeline successfully uploaded the WAR file to Nexus.

The console confirms:

Artifact: hiring.war

GroupId: com.hiring

ArtifactId: hiring

Version: 1.0

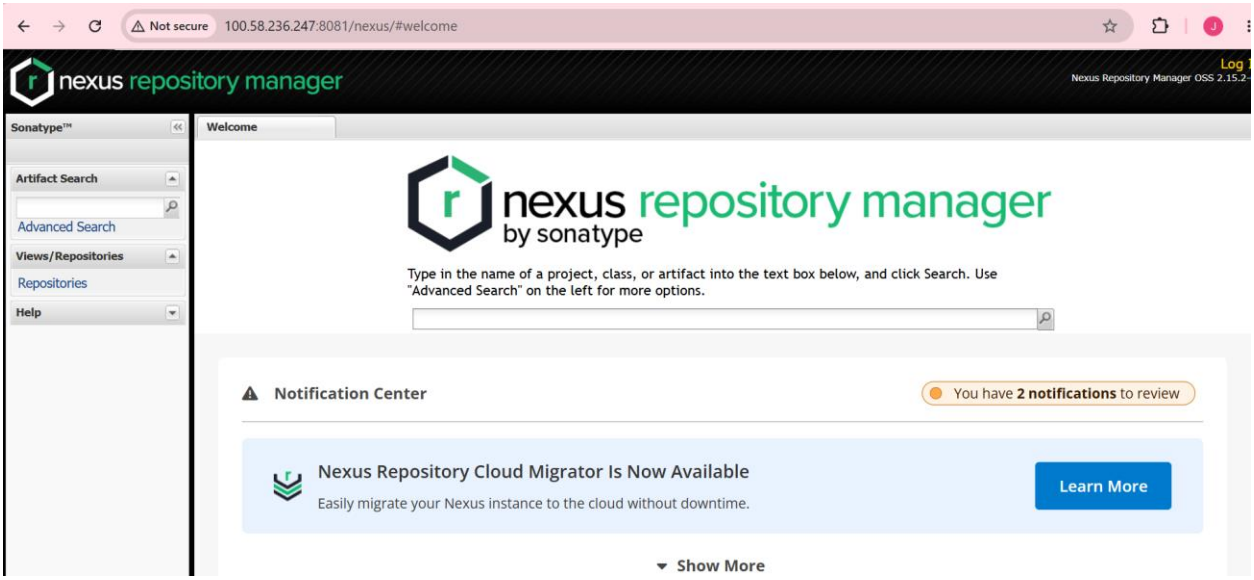
Repository: releases

Upload: 100% completed

Final status: Finished: SUCCESS

So your flow is now:

GitHub → Jenkins → Maven Build → hiring.war → Nexus Repository ✅





Releases → com → hiring → hiring → 1.0

And inside 1.0:

✅ **hiring-1.0.war**

hiring-1.0.war.md5

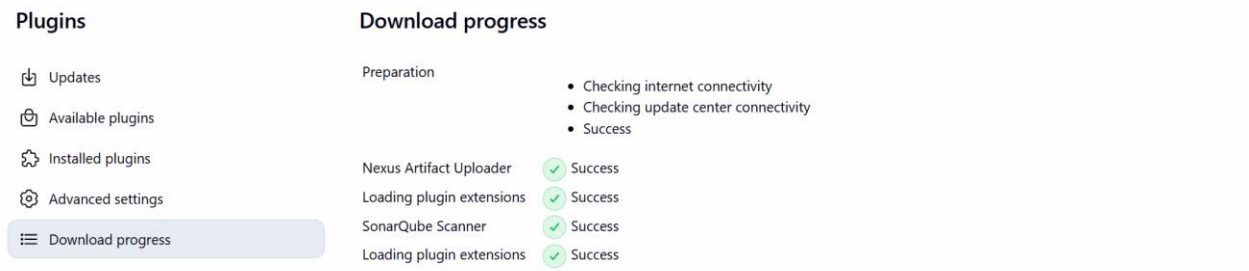
hiring-1.0.war.sha1

So your complete flow is working:

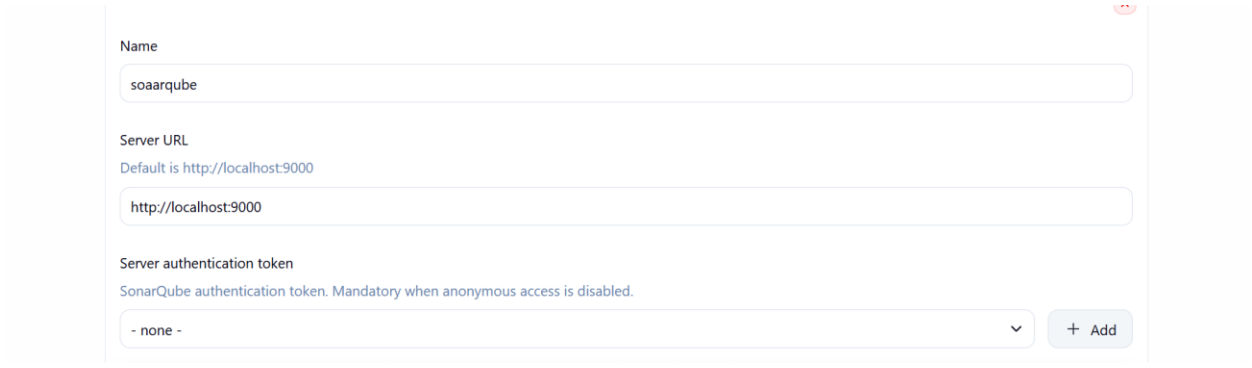
GitHub → Jenkins → Maven → WAR → Nexus Releases Repository ✅

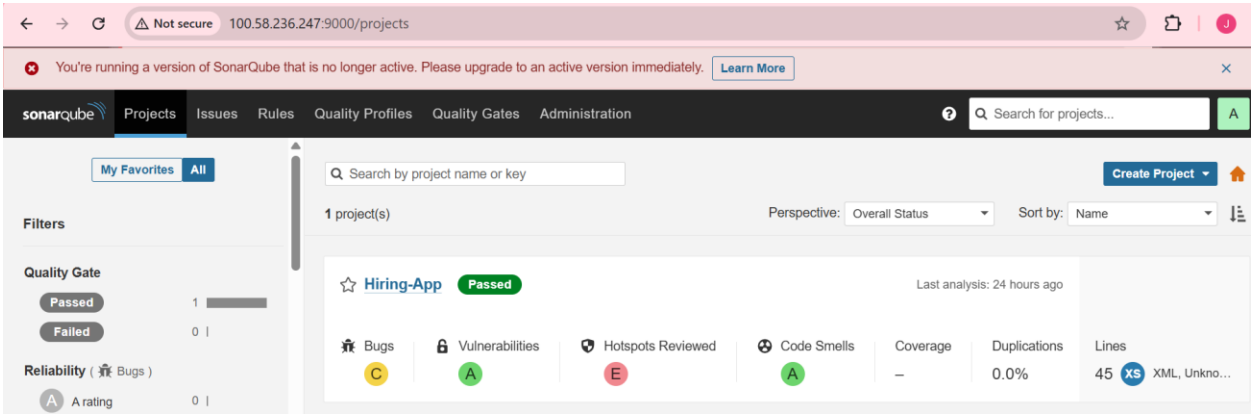
This Nexus upload task is 100% completed.

Now confiure sonarqube with jenkins

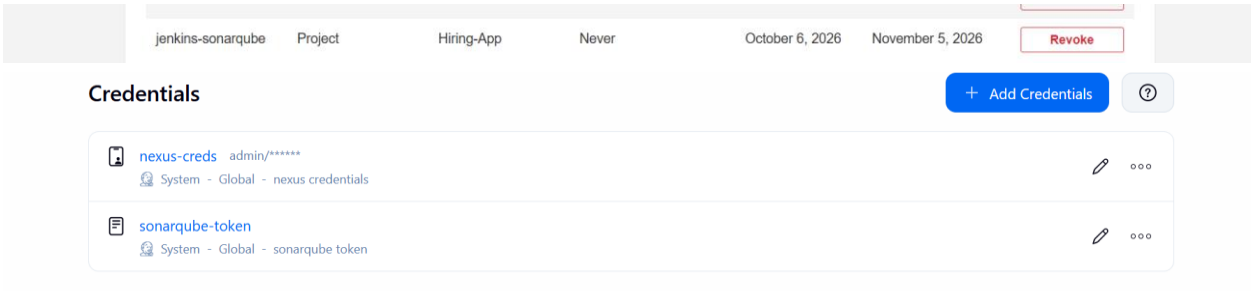


Downloaded sonarqube scanner

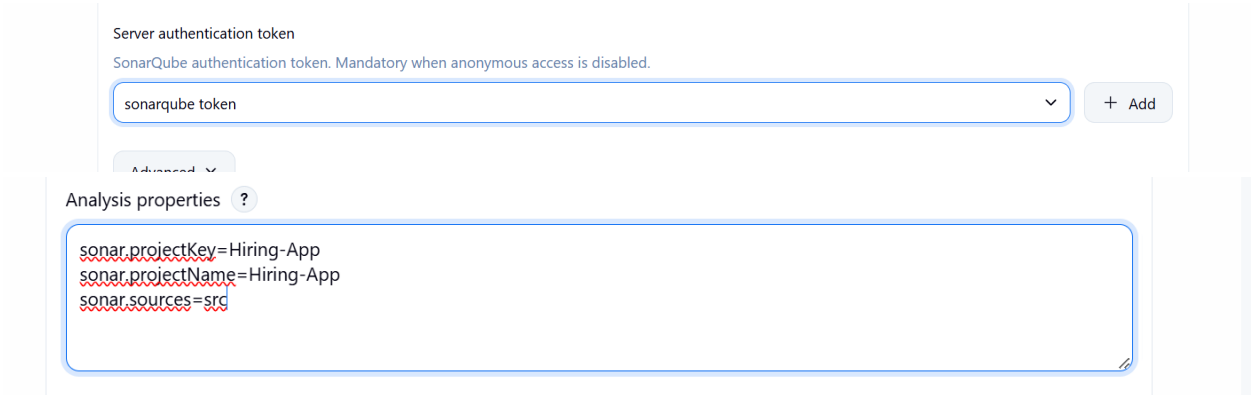




Create new token in sonarqube

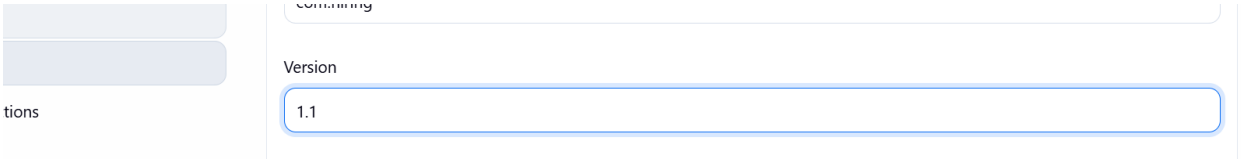


Add the credentials

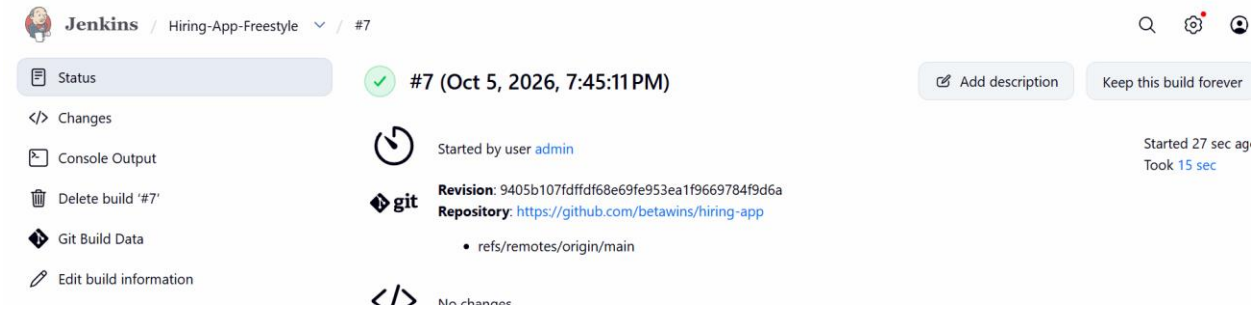


```
Failed to deploy artifacts: Could not transfer artifact com.hiring:hiring:war:1.0 from/to releases
(http://localhost:8081/nexus/content/repositories/releases): transfer failed for
http://localhost:8081/nexus/content/repositories/releases/com/hiring/hiring/1.0/hiring-1.0.war, status: 400
Bad Request
FATAL: Failed to deploy artifacts: Could not transfer artifact com.hiring:hiring:war:1.0 from/to releases
(http://localhost:8081/nexus/content/repositories/releases): transfer failed for
http://localhost:8081/nexus/content/repositories/releases/com/hiring/hiring/1.0/hiring-1.0.war, status: 400
Bad Request
java.io.IOException: Failed to deploy artifacts: Could not transfer artifact com.hiring:hiring:war:1.0
from/to releases (http://localhost:8081/nexus/content/repositories/releases): transfer failed for
http://localhost:8081/nexus/content/repositories/releases/com/hiring/hiring/1.0/hiring-1.0.war, status: 400
Bad Request
```

Failure



Change version to 1.1



Build is success

SonarQube analysis: ANALYSIS SUCCESSFUL

Execution: SUCCESS

Jenkins: Finished: SUCCESS ✅

So the Jenkins → SonarQube integration is working correctly.

GitHub → Jenkins → Maven Build → SonarQube → Nexus ✅

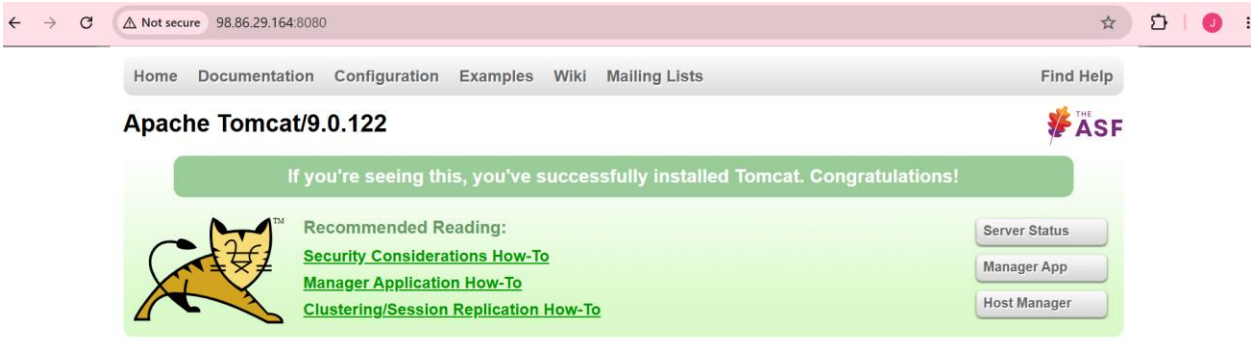
Launch tomcat-server new instance

```
[root@ip-10-0-11-23 ~]# java --version
openjdk 17.0.20.1 2026-08-18 LTS
OpenJDK Runtime Environment Corretto-17.0.20.10.1 (build 17.0.20.1+10-LTS)
OpenJDK 64-Bit Server VM Corretto-17.0.20.10.1 (build 17.0.20.1+10-LTS, mixed mode, sharing)
[root@ip-10-0-11-23 ~]#
```

```
Using CATALINA_OPTS:
Tomcat started.
[ec2-user@ip-10-0-11-23 bin]$ curl http://localhost:8080



<!DOCTYPE html>
<html lang="en">
    <head>
        <meta charset="UTF-8" />
        <title>Apache Tomcat/9.0.122</title>
        <link href="favicon.ico" rel="icon" type="image/x-icon" />
        <link href="tomcat.css" rel="stylesheet" type="text/css" />
```




```
[ec2-user@ip-10-0-11-23 bin]$ useradd -r -s /bin/false tomcat
useradd: Permission denied.
useradd: cannot lock /etc/passwd; try again later.
[ec2-user@ip-10-0-11-23 bin]$ sudo useradd -r -s /bin/false tomcat
[ec2-user@ip-10-0-11-23 bin]$ chown -R tomcat:tomcat /opt/tomcat
```

We already have the WAR in Nexus:

Nexus → releases → com → hiring → hiring → 1.0 → hiring-1.0.war

Now we'll download it to the Tomcat server and deploy it.

```
[ec2-user@ip-10-0-11-23 webapps]$ wget --user=admin --password='admin123' -O hiring.war http://100.58.236.247:8081/nexus/service/local/repos
itories/releases/content/com/hiring/hiring/1.0/hiring-1.0.war
--2026-10-06 07:04:12--  http://100.58.236.247:8081/nexus/service/local/repositories/releases/content/com/hiring/hiring/1.0/hiring-1.0.war
Connecting to 100.58.236.247:8081... connected.
HTTP request sent, awaiting response... 200 OK
Length: 1698 (1.7K) [application/java-archive]
Saving to: 'hiring.war'

hiring.war                      100%[=============================================================>]   1.66K  --.-KB/s    in 0s

2026-10-06 07:04:12 (304 MB/s) - 'hiring.war' saved [1698/1698]
```

wget → download a file from a URL

--user=admin → Nexus username

--password='admin123' → Nexus password

-O hiring.war → save the downloaded file as hiring.war

100.58.236.247:8081 → Nexus server

/nexus/... → location of the WAR inside Nexus

hiring-1.0.war → the application WAR we built

```
[ec2-user@ip-10-0-11-23 webapps]$ ls
ROOT  docs  examples  hiring  hiring.war  host-manager  manager
[ec2-user@ip-10-0-11-23 webapps]$ cd /opt/tomcat/bin
[ec2-user@ip-10-0-11-23 bin]$ ./startup.sh
Using CATALINA_BASE:   /opt/tomcat
Using CATALINA_HOME:   /opt/tomcat
Using CATALINA_TMPDIR: /opt/tomcat/temp
Using JRE_HOME:        /usr
Using CLASSPATH:       /opt/tomcat/bin/bootstrap.jar:/opt/tomcat/bin/tomcat-juli.jar
Using CATALINA_OPTS:
Tomcat started.
[ec2-user@ip-10-0-11-23 bin]$ ls -l /opt/tomcat/webapps/
total 36
drwxr-x---.  3 ec2-user ec2-user 16384 Oct  6 06:36 ROOT
drwxr-x---. 16 ec2-user ec2-user 16384 Oct  6 06:36 docs
drwxr-x---.  7 ec2-user ec2-user    99 Oct  6 06:36 examples
drwxr-x---.  4 ec2-user ec2-user    54 Oct  6 07:04 hiring
-rw-r--r--.  1 ec2-user ec2-user  1698 Oct  5 19:17 hiring.war
drwxr-x---.  6 ec2-user ec2-user    79 Oct  6 06:36 host-manager
drwxr-x---.  6 ec2-user ec2-user   114 Oct  6 06:36 manager
[ec2-user@ip-10-0-11-23 bin]$
```



Welcome to Techie Horizon

GitHub → Jenkins → Maven → WAR → Nexus → Tomcat → Browser ✅

```
07:16:04.779 INFO  Analysis report uploaded in 78ms
07:16:04.785 INFO  ANALYSIS SUCCESSFUL, you can find the results at: http://localhost:9000/dashboard?id=Hiring-App
07:16:04.785 INFO  Note that you will be able to access the updated dashboard once the server has processed the submitted analysis report
07:16:04.785 INFO  More about the report processing at http://localhost:9000/api/ce/task?id=AaEQEc4WxqMr57LQ3dJu
07:16:04.808 INFO  Analysis total time: 6.163 s
07:16:04.812 INFO  EXECUTION SUCCESS
07:16:04.813 INFO  Total time: 9.189s
Finished: SUCCESS
```

2) Setup a Jenkins CICD pipeline using Declarative pipeline using feature-1.1 branch.

Source Code: **https://github.com/betawins/sabear_simplecutomerapp/tree/feature-1.1**

stages:

- **Git Clone**
- **SonarQube Integration**
- **Maven Compilation**
- **Nexus Artifactory**

- **Slack Notification**
- **Deploy On tomcat**




```
[root@ip-10-0-6-42 ~]# git ls-remote https://github.com/betawins/sabear_simplecutomerapp.git
9be96d5661cf61f49c0b8d076dc20d2b09624f4e        HEAD
05222fef4b498f40dd73a8a1b6fbdb0c619d398a        refs/heads/feature-1.1
9be96d5661cf61f49c0b8d076dc20d2b09624f4e        refs/heads/master
3686a94fa32cd222b53ad48d57c7561e1cc99759        refs/pull/1/head
b24eb28ca1ffc7b5f2f65c2aa3c7f5e3985d5a38        refs/pull/10/head
c4702b90000347ff4e2c996e65aebf0ed972bb2a        refs/pull/11/head
f80ec65acbf755900b31225f245164377ea4e89f        refs/pull/12/head
ffc35279f42d062dbdf996b3a1570152713cdb9f        refs/pull/12/merge
1cea44ffc09aa8e54bd105010c85aba7d01ca2d8        refs/pull/13/head
9ae458c9643eaa512435e5bd08d0d57ef7315440        refs/pull/14/head
692f26a54927cf78559872ba17b3dc38c507d347        refs/pull/15/head
9fc0760d03c9061419426a805293b5f55b9aa86e        refs/pull/15/merge
74c9ccf09a706531b06ccea961340982155c5505        refs/pull/2/head
```

```
[root@ip-10-0-6-42 ~]# git ls-remote https://github.com/betawins/sabear_simplecutomerapp.git refs/heads/feature-1.1
05222fef4b498f40dd73a8a1b6fbdb0c619d398a        refs/heads/feature-1.1
```

```
pipeline {
    agent any

    stages {
        stage('Git Clone') {
            steps {
                git branch: 'feature-1.1',
                    url: 'https://github.com/betawins/sabear_simplecutomerapp.git'
            }
        }
    }
}
```

Save the script

Then build now, the build is successful

```
Complete!
[root@ip-10-0-13-220 ~]# useradd -m -s /bin/bash sonar
[root@ip-10-0-13-220 ~]#
```

Add user for sonar and install sonarqube

```
[root@ip-10-0-13-220 ~]# cd /opt
wget https://binaries.sonarsource.com/Distribution/sonarqube/sonarqube-26.1.0.118079.zip
--2026-10-06 09:47:13--  https://binaries.sonarsource.com/Distribution/sonarqube/sonarqube-26.1.0.118079.zip
Resolving binaries.sonarsource.com (binaries.sonarsource.com)... 3.167.37.10, 3.167.37.121, 3.167.37.88, ...
Connecting to binaries.sonarsource.com (binaries.sonarsource.com)|3.167.37.10|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 909320623 (867M) [binary/octet-stream]
Saving to: 'sonarqube-26.1.0.118079.zip'

sonarqube-26.1.0.118079.zip        100%[=============================================================>] 867.20M   398MB/

2026-10-06 09:47:16 (398 MB/s) - 'sonarqube-26.1.0.118079.zip' saved [909320623/909320623]

[root@ip-10-0-13-220 opt]# cd /opt
unzip sonarqube-26.1.0.118079.zip
```

Extract it

```
[root@ip-10-0-13-220 opt]# mv /opt/sonarqube-26.1.0.118079 /opt/sonarqube
[root@ip-10-0-13-220 opt]# chown -R sonar:sonar /opt/sonarqube
[root@ip-10-0-13-220 opt]#
```

Change name and give permissions

```
[root@ip-10-0-13-220 opt]# nano /etc/systemd/system/sonarqube.service
[root@ip-10-0-13-220 opt]#
```

[Unit]

Description=SonarQube service

After=network.target

[Service]

Type=forking

User=sonar

Group=sonar

ExecStart=/opt/sonarqube/bin/linux-x86-64/sonar.sh start

ExecStop=/opt/sonarqube/bin/linux-x86-64/sonar.sh stop

Restart=always

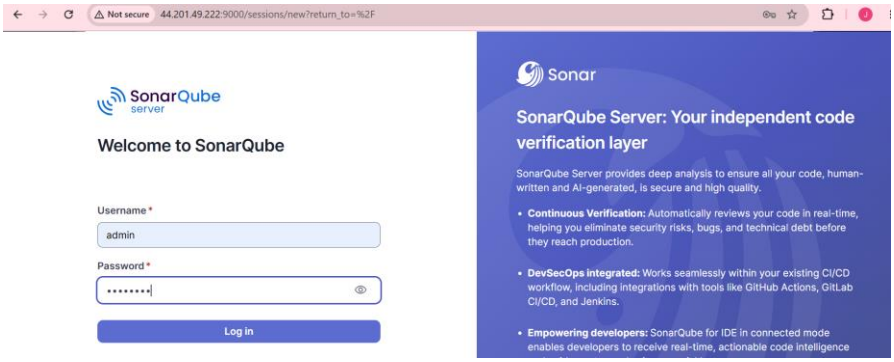
[Install]

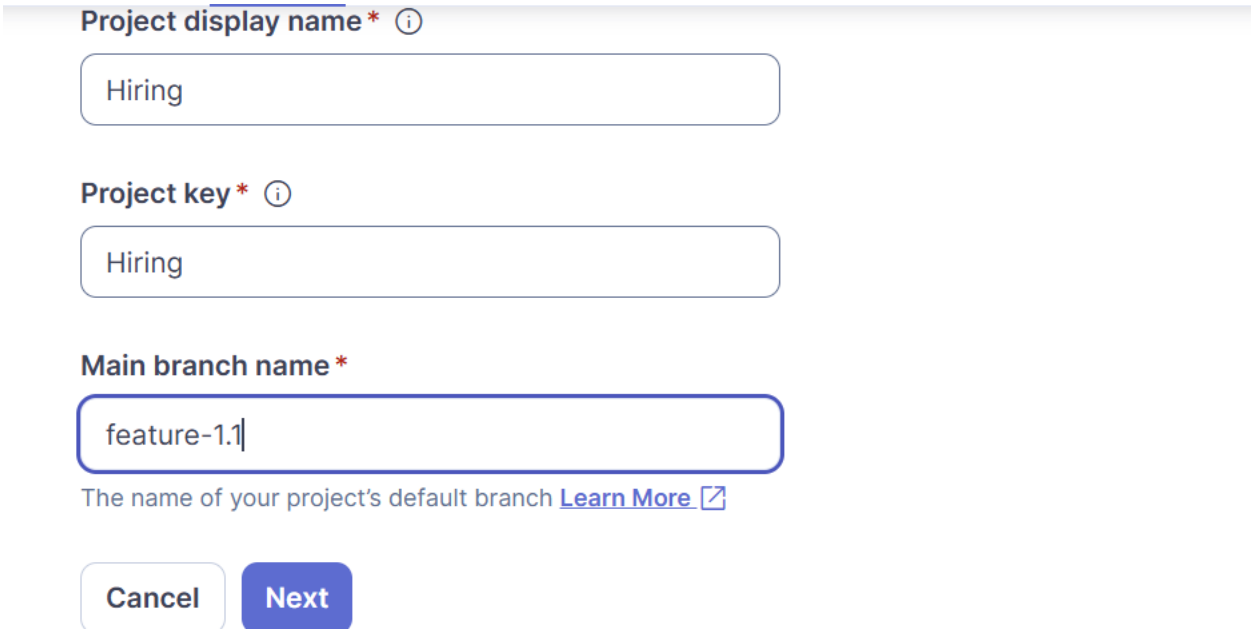
WantedBy=multi-user.target

Paste this script

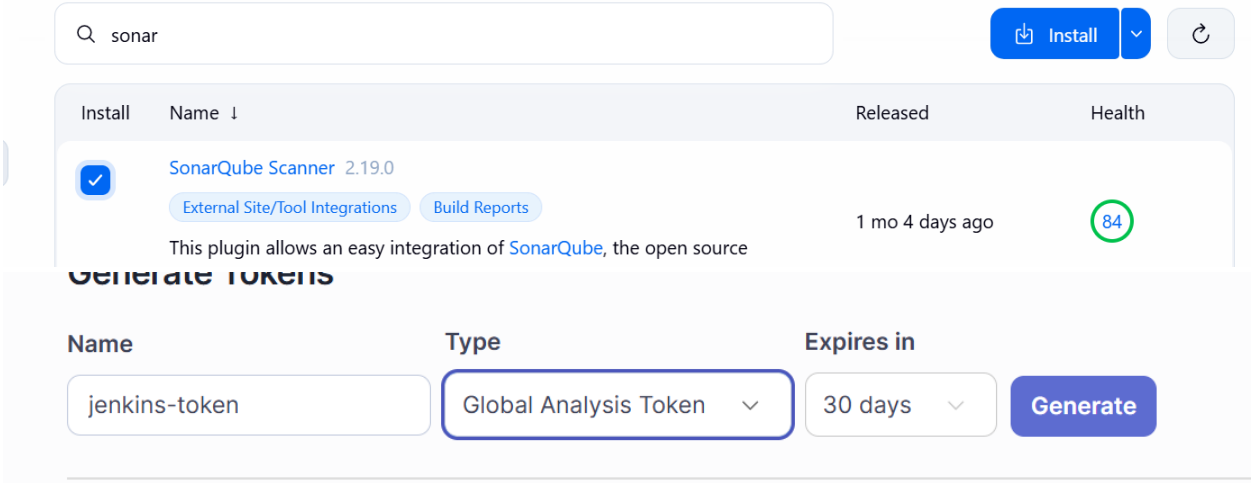
```
[root@ip-10-0-13-220 opt]# systemctl daemon-reload
[root@ip-10-0-13-220 opt]# systemctl enable sonarqube
Created symlink /etc/systemd/system/multi-user.target.wants/sonarqube.service → /etc/systemd/system/sonarqube.service.
[root@ip-10-0-13-220 opt]# systemctl start sonarqube
[root@ip-10-0-13-220 opt]# systemctl status sonarqube
● sonarqube.service - SonarQube service
     Loaded: loaded (/etc/systemd/system/sonarqube.service; enabled; preset: disabled)
```

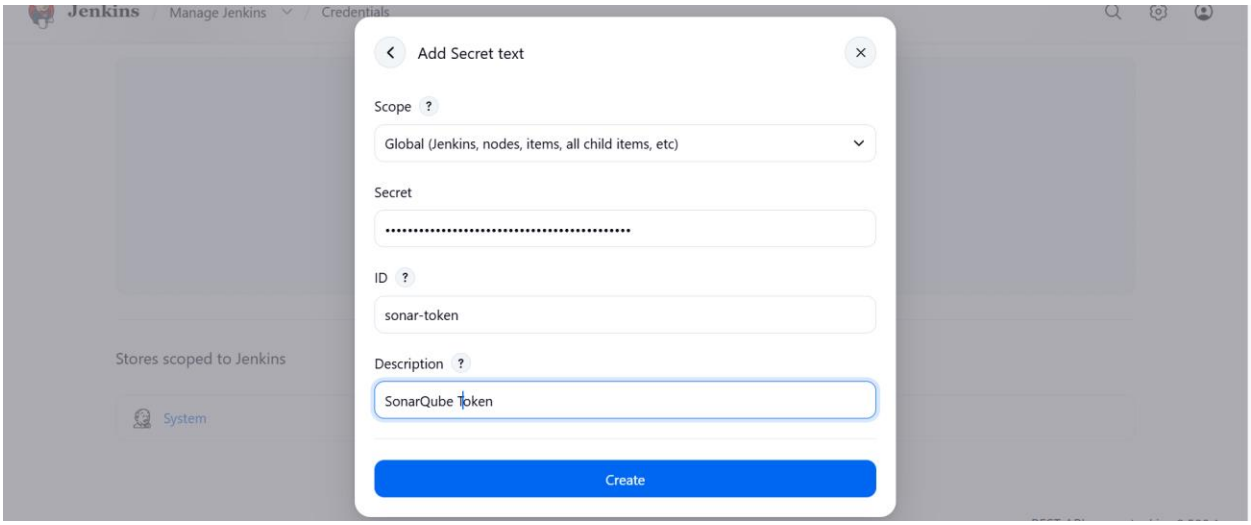
Start the sonarqube





Create project






```
[root@ip-10-0-13-220 opt]# dnf install maven -y
Last metadata expiration check: 0:21:38 ago on Tue Oct  6 09:45:10 2026.
Dependencies resolved.
================================================================================
 Package                 Architecture       Version                 Repository
================================================================================
Installing:
 maven                   noarch             1:3.8.4-3.amzn2023.0.5  amazonlinux
Installing dependencies:
```

Install maven

```
[root@ip-10-0-13-220 opt]# mvn -version
Apache Maven 3.8.4 (Red Hat 3.8.4-3.amzn2023.0.5)
Maven home: /usr/share/maven
Java version: 17.0.20.1, vendor: Amazon.com Inc., runtime: /usr/lib/jvm/java-17-amazon-corretto.x86_64
Default locale: en, platform encoding: UTF-8
```

try sample Pipelin

```
4    stages {
5
6        stage('Git Clone') {
7            steps {
8                git branch: 'feature-1.1',
9                    url: 'https://github.com/betawins/sabear_simplecutomerapp.git'
10           }
11       }
12
13       stage('SonarQube Integration') {
14           steps {
15               withSonarQubeEnv('SonarQube') {
16                   sh 'mvn clean verify sonar:sonar -Dsonar.projectKey=Hiring'
17               }
```

```
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  23.547 s
[INFO] Finished at: 2026-10-06T10:59:50Z
[INFO] ------------------------------------------------------------------------
[Pipeline] }
[Pipeline] // withSonarQubeEnv
[Pipeline] }
[Pipeline] // stage
[Pipeline] }
[Pipeline] // node
[Pipeline] End of Pipeline
Finished: SUCCESS
```

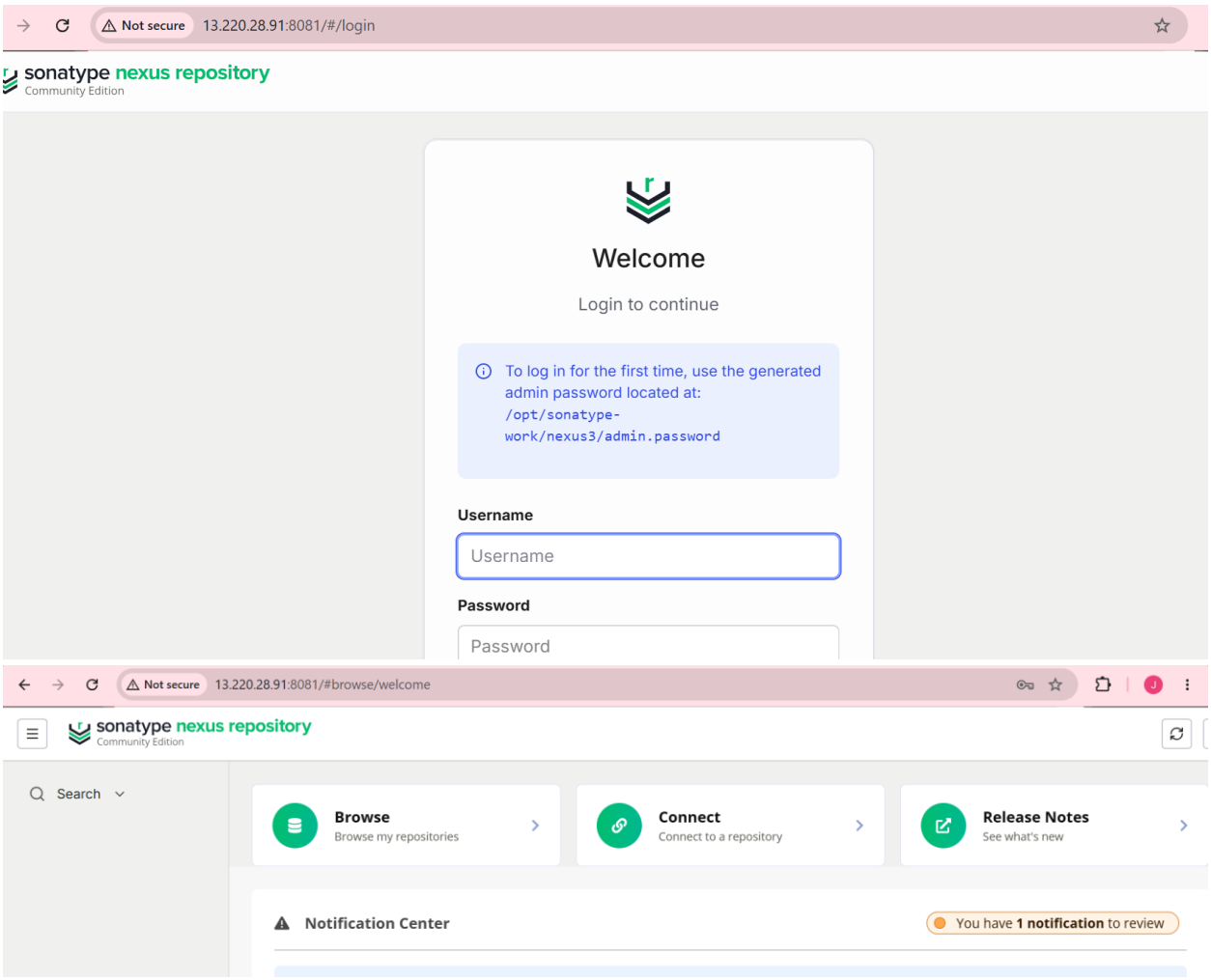
```
Complete!
[root@ip-10-0-6-223 ~]# java --version
openjdk 21.0.12.1 2026-08-18 LTS
OpenJDK Runtime Environment Corretto-21.0.12.9.1 (build 21.0.12.1+9-LTS)
OpenJDK 64-Bit Server VM Corretto-21.0.12.9.1 (build 21.0.12.1+9-LTS, mixed mode, sharing)
```

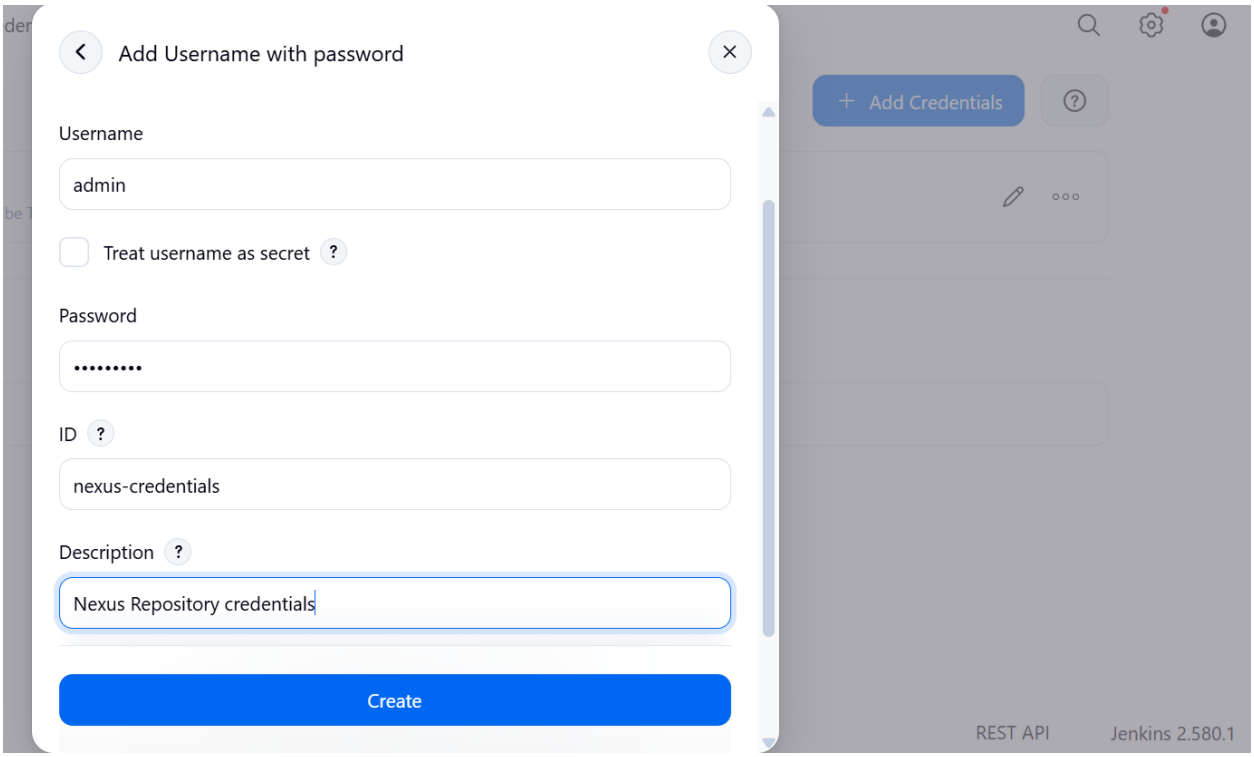
```
[root@ip-10-0-6-223 opt]# sudo wget https://download.sonatype.com/nexus/3/nexus-3.96.3-01-linux-x86_64.tar.gz
```

```
[root@ip-10-0-6-223 opt]# sudo tar -xzf nexus-3.96.3-01-linux-x86_64.tar.gz
[root@ip-10-0-6-223 opt]# sudo chown -R nexus:nexus /opt/nexus-3.96.3-01
[root@ip-10-0-6-223 opt]# sudo sed -i 's/^#run_as_user=.*/run_as_user="nexus"/' /opt/nexus-3.96.3-01/bin/nexus.rc
sed: can't read /opt/nexus-3.96.3-01/bin/nexus.rc: No such file or directory
[root@ip-10-0-6-223 opt]# ls -l /opt/nexus-3.96.3-01/bin/
total 327652
-rwxr-xr-x. 1 nexus nexus      6735 Sep 21 22:51 nexus
-rw-r--r--. 1 nexus nexus      1325 Sep 21 22:51 nexus.vmoptions
-rw-r--r--. 1 nexus nexus 335499921 Sep 21 23:47 sonatype-nexus-repository-3.96.3-01.jar
[root@ip-10-0-6-223 opt]#
```

```
[root@ip-10-0-6-223 opt]# sudo mkdir -p /opt/sonatype-work/nexus3
sudo chown -R nexus:nexus /opt/sonatype-work
[root@ip-10-0-6-223 opt]# sudo -u nexus /opt/nexus-3.96.3-01/bin/nexus start
Starting nexus
[root@ip-10-0-6-223 opt]#
```

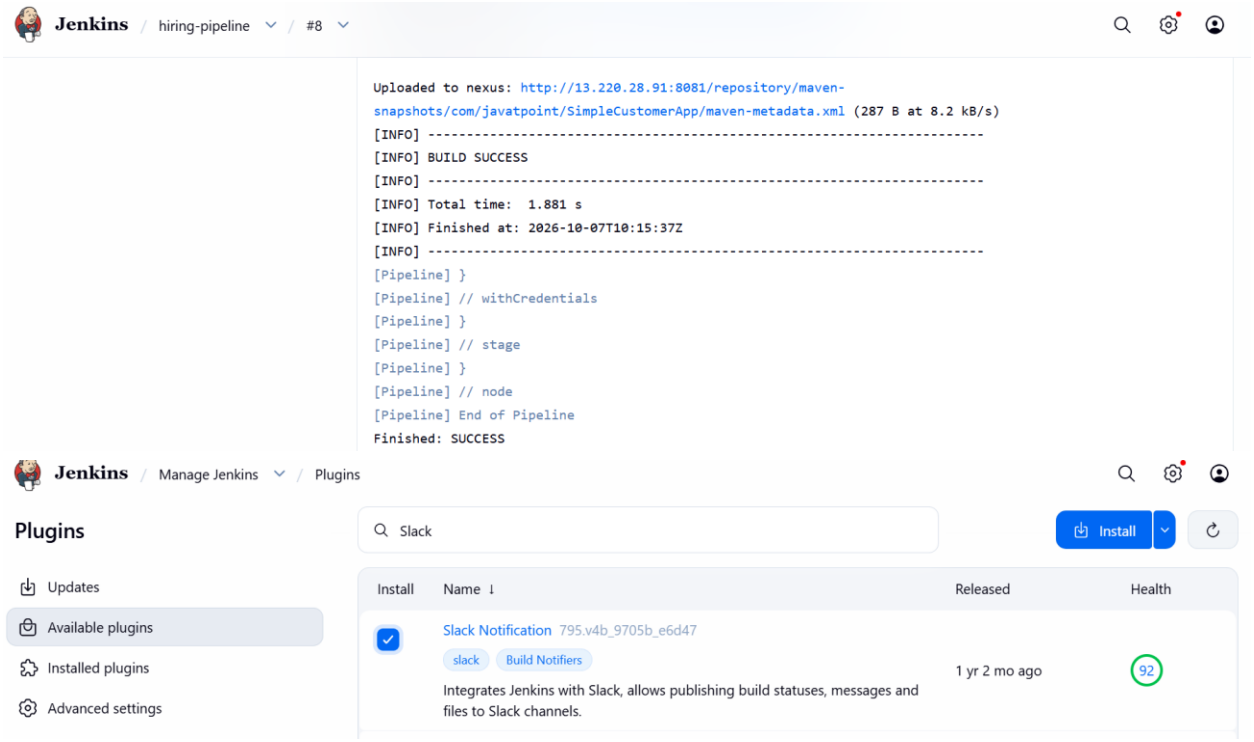
```
[root@ip-10-0-6-223 opt]# sudo -u nexus /opt/nexus-3.96.3-01/bin/nexus status
[root@ip-10-0-6-223 opt]# ss -lntp | grep 8081
LISTEN 0      50                 *:8081             *:*    users:(("java",pid=28106,fd=18))
[root@ip-10-0-6-223 opt]#
```


Not secure
13.220.28.91:8081/#/login
sonatype nexus repository
Community Edition
Welcome
Login to continue
To log in for the first time, use the generated admin password located at:
/opt/sonatype-work/nexus3/admin.password
Username
Username
Password
Password
Not secure
13.220.28.91:8081/#browse/welcome
sonatype nexus repository
Community Edition
Search
Browse
Browse my repositories
Connect
Connect to a repository
Release Notes
See what's new
Notification Center
You have 1 notification to review



Add credentials in the jenkins

Add the new script related to nexus in the jenkins configure

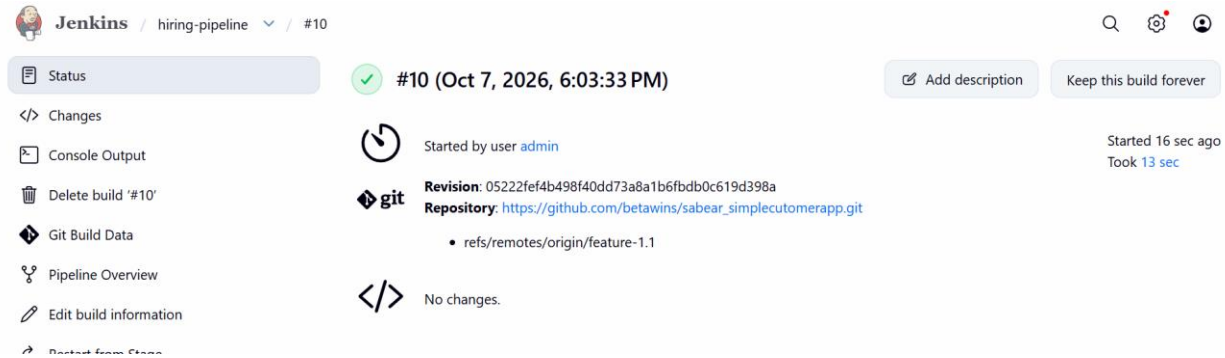


Install slack notification

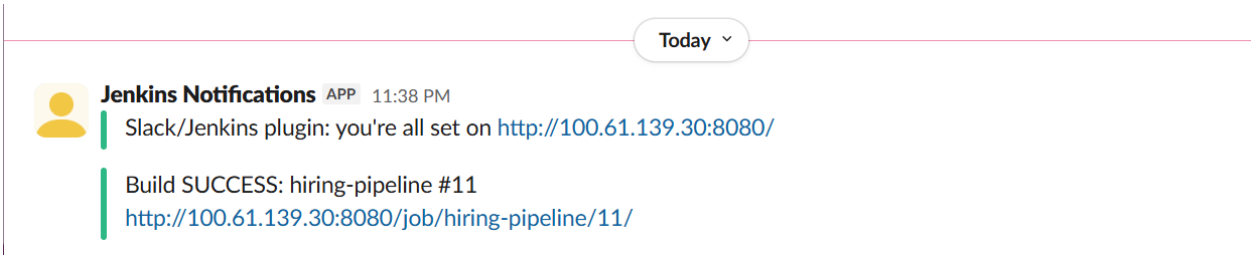
```
}
stage('Slack Notification') {
    steps {
        slackSend(
            color: 'good',
            message: "Build SUCCESS: ${env.JOB_NAME} #${env.BUILD_NUMBER}\n${env.BUILD_URL}"
        )
    }
}
```

Use Groovy Sandbox ?

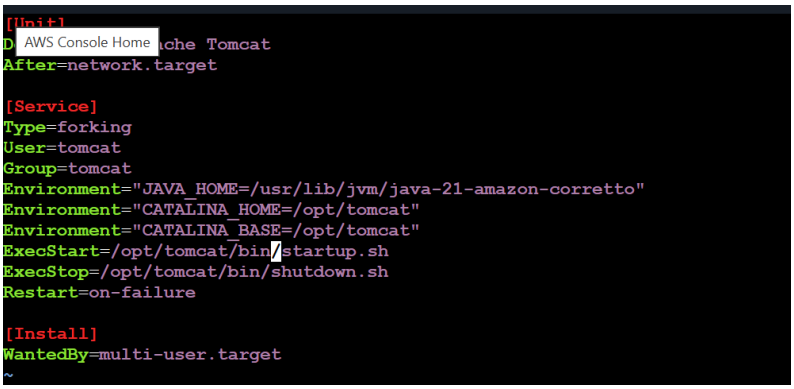
Use this script and save



Success



Install tomcat



Add this in the service file

```
++ basename target/SimpleCustomerApp-13-SNAPSHOT.war
+ WAR_NAME=SimpleCustomerApp-13-SNAPSHOT.war
+ scp -i **** -o StrictHostKeyChecking=no target/SimpleCustomerApp-13-SNAPSHOT.war ec2-
user@100.29.189.251:/tmp/SimpleCustomerApp-13-SNAPSHOT.war
Warning: Permanently added '100.29.189.251' (ED25519) to the list of known hosts.
+ ssh -i **** -o StrictHostKeyChecking=no ec2-user@100.29.189.251 'sudo cp /tmp/SimpleCustomerApp-13-
SNAPSHOT.war /opt/tomcat/webapps/SimpleCustomerApp-13-SNAPSHOT.war && sudo chown tomcat:tomcat
/opt/tomcat/webapps/SimpleCustomerApp-13-SNAPSHOT.war'
[Pipeline] }
[Pipeline] // withCredentials
[Pipeline] }
[Pipeline] // stage
[Pipeline] }
[Pipeline] // node
[Pipeline] End of Pipeline
Finished: SUCCESS
```

Deploying in the tomcat is done

```
[INFO] Packaging webapp
[INFO] Assembling webapp [SimpleCustomerApp] in [/var/lib/jenkins/workspace/hiring-pipeline/target/SimpleCustomerApp-${BUILD
T]
[INFO] Processing war project
[INFO] Copying webapp resources [/var/lib/jenkins/workspace/hiring-pipeline/WebContent]
[INFO] Building war: /var/lib/jenkins/workspace/hiring-pipeline/target/SimpleCustomerApp-${BUILD_NUMBER}-SNAPSHOT.war
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  29.002 s
[INFO] Finished at: 2026-10-07T19:48:14Z
[INFO] ------------------------------------------------------------------------
```

Maven is packaging the webcontent in to the war

Your pipeline flow is now

GitHub feature-1.1

↓

Git Clone

↓

Fix WAR packaging (WebContent)

↓

SonarQube analysis

↓

Maven build

↓

Upload to Nexus

↓

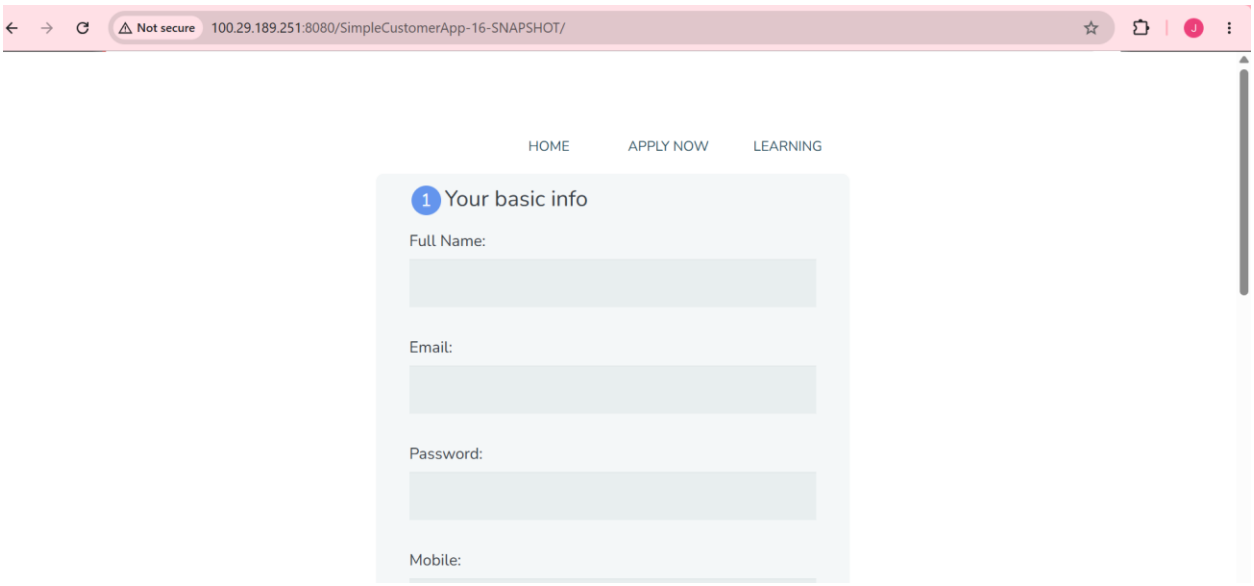
Slack notification

↓

Deploy WAR to Tomcat

Build 16 is success so run on the browser using

http://100.29.189.251:8080/SimpleCustomerApp-16-SNAPSHOT/



3) Setup a Jenkins CICD pipeline using Scripted pipeline using feature-1.1 branch.
Source Code: **https://github.com/betawins/sabear_simplecutomerapp/tree/feature-1.1**
stages:

- **Git Clone**
- **SonarQube Integration**
- **Maven Compilation**
- **Nexus Artifactory**
- **Slack Notification**
- **Deploy On tomcat**



Use this script in the configure pipeline

```
node {

   stage('Git Clone') {
      git branch: 'feature-1.1',
         url: 'https://github.com/betawins/sabear_simplecutomerapp.git'

      // Make sure Maven packages WebContent into the WAR
      sh '''
```

```
        sed -i '/<failOnMissingWebXml>false<\\/failOnMissingWebXml>/a\\
        <warSourceDirectory>WebContent</warSourceDirectory>' pom.xml
      '''
    }

    stage('Maven Compilation') {
      sh 'mvn clean package -DskipTests'
    }

    stage('SonarQube Integration') {
      withSonarQubeEnv('SonarQube') {
        sh '''
          mvn org.sonarsource.scanner.maven:sonar-maven-plugin:sonar \
          -Dsonar.projectKey=Hiring
        '''
      }
    }

    stage('Nexus Artifactory') {
      withCredentials([usernamePassword(
        credentialsId: 'nexus-credentials',
        usernameVariable: 'NEXUS_USER',
        passwordVariable: 'NEXUS_PASS'
      )]) {

        sh '''
          cat > settings.xml <<EOF
<settings>
 <servers>
  <server>
   <id>nexus</id>
   <username>${NEXUS_USER}</username>
   <password>${NEXUS_PASS}</password>
  </server>
 </servers>
</settings>
EOF

          mvn deploy \
           -s settings.xml \
           -DaltDeploymentRepository=nexus::default::http://13.220.28.91:8081/repository/maven-snapshots/
```

```
            '''
        }
    }

    stage('Slack Notification') {
        slackSend(
            color: 'good',
            message: "Build SUCCESS: ${env.JOB_NAME} #${env.BUILD_NUMBER}\n${env.BUILD_URL}"
        )
    }

    stage('Deploy On Tomcat') {
        withCredentials([sshUserPrivateKey(
            credentialsId: 'tomcat-ssh',
            keyFileVariable: 'SSH_KEY',
            usernameVariable: 'SSH_USER'
        )]) {

            sh '''
                WAR=$(ls target/*.war | head -1)
                WAR_NAME=$(basename "$WAR")

                scp -i "$SSH_KEY" \
                    -o StrictHostKeyChecking=no \
                    "$WAR" "$SSH_USER@100.29.189.251:/tmp/$WAR_NAME"

                ssh -i "$SSH_KEY" \
                    -o StrictHostKeyChecking=no \
                    "$SSH_USER@100.29.189.251" \
                    "sudo rm -rf /opt/tomcat/webapps/SimpleCustomerApp-* && \
                     sudo cp /tmp/$WAR_NAME /opt/tomcat/webapps/$WAR_NAME && \
                     sudo chown tomcat:tomcat /opt/tomcat/webapps/$WAR_NAME"
            '''
        }
    }
}
```

```
++ basename target/SimpleCustomerApp-1-SNAPSHOT.war
+ WAR_NAME=SimpleCustomerApp-1-SNAPSHOT.war
+ scp -i **** -o StrictHostKeyChecking=no target/SimpleCustomerApp-1-SNAPSHOT.war ec2-
user@100.29.189.251:/tmp/SimpleCustomerApp-1-SNAPSHOT.war
+ ssh -i **** -o StrictHostKeyChecking=no ec2-user@100.29.189.251 'sudo rm -rf
/opt/tomcat/webapps/SimpleCustomerApp-* &&                    sudo cp /tmp/SimpleCustomerApp-1-SNAPSHOT.war
/opt/tomcat/webapps/SimpleCustomerApp-1-SNAPSHOT.war &&                    sudo chown tomcat:tomcat
/opt/tomcat/webapps/SimpleCustomerApp-1-SNAPSHOT.war'
[Pipeline] }
[Pipeline] // withCredentials
[Pipeline] }
[Pipeline] // stage
[Pipeline] }
[Pipeline] // node
[Pipeline] End of Pipeline
Finished: SUCCESS
```

The build was succcessfull



4) Write sample skeleton of pipelines.



Create new job

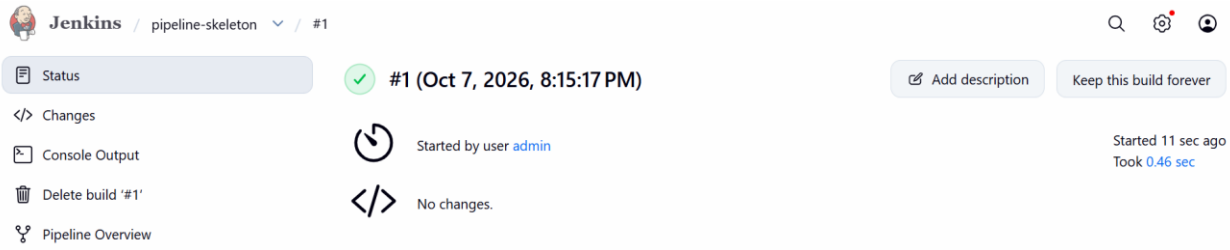
Use this script;

```
pipeline {
  agent any

  stages {
    stage('Build') {
      steps {
        echo 'Build stage'
      }
    }

    stage('Test') {
      steps {
        echo 'Test stage'
      }
    }
```

```
    stage('Deploy') {
      steps {
        echo 'Deploy stage'
      }
    }
  }
}
```



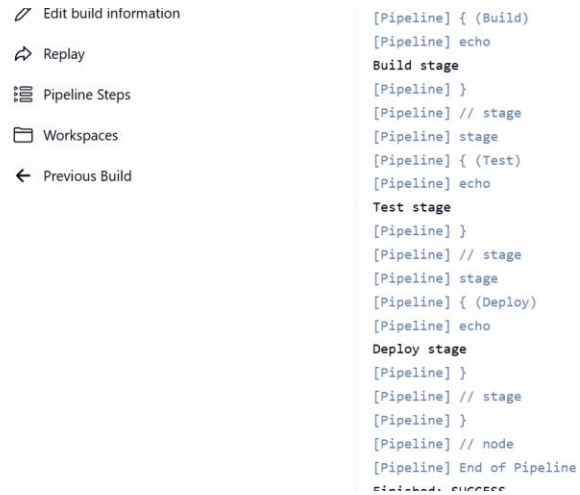
Replace the script by;

```
node {

  stage('Build') {
    echo 'Build stage'
  }

  stage('Test') {
    echo 'Test stage'
  }

  stage('Deploy') {
    echo 'Deploy stage'
  }
}
```

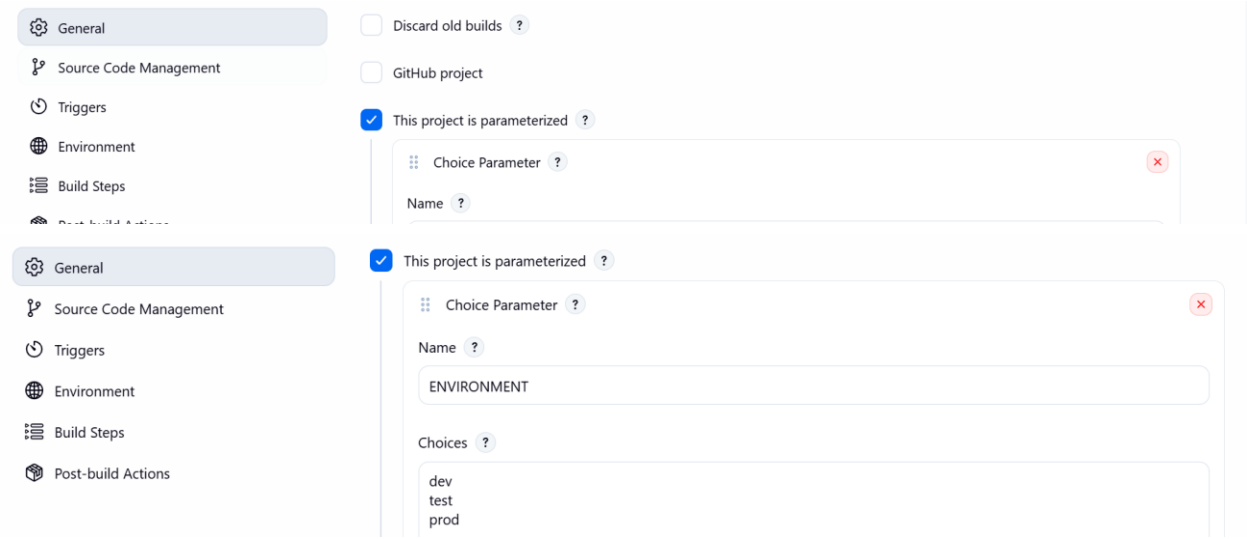


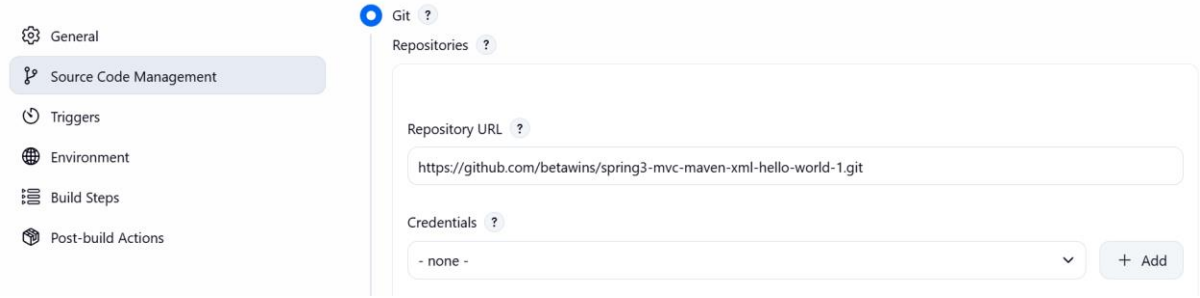
5) Create a parameterized job in Jenkins.

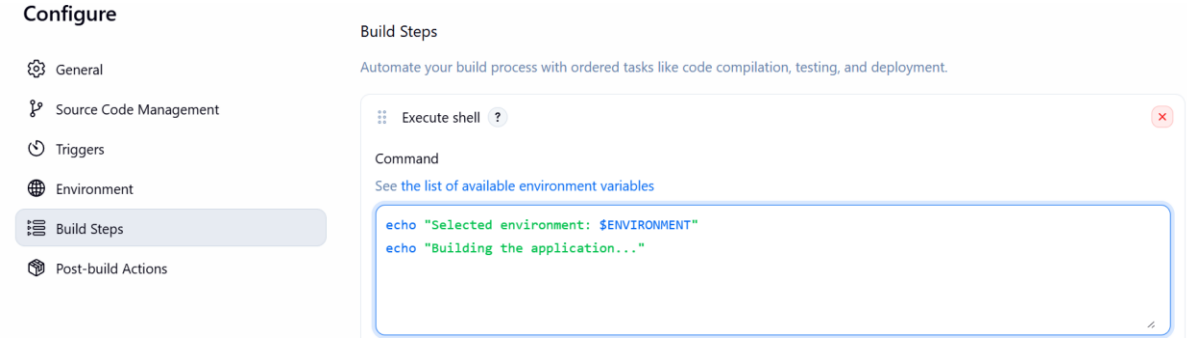
Source Code: https://github.com/betawins/spring3-mvc-maven-xml-hello-world-1.git

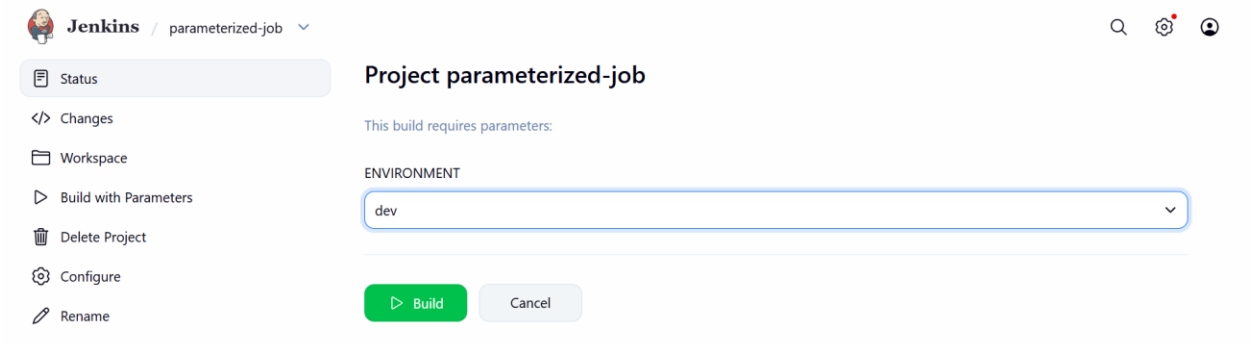


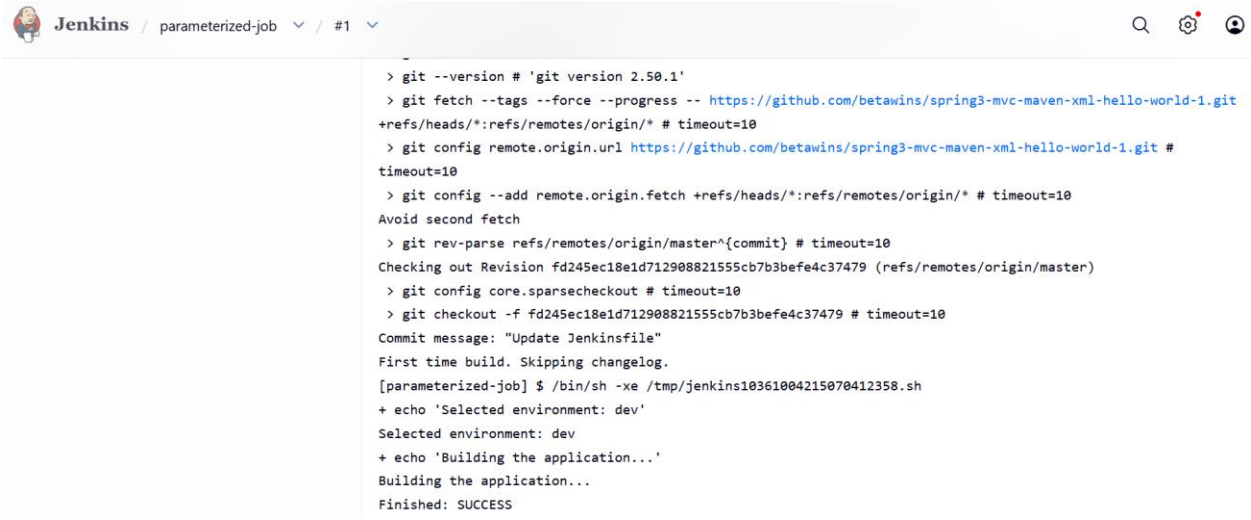
Create new job using freestyle project











6) Setup one slave machine for Jenkins.

```
[root@ip-10-0-15-183 ~]# java --version
openjdk 21.0.12.1 2026-08-18 LTS
OpenJDK Runtime Environment Corretto-21.0.12.9.1 (build 21.0.12.1+9-LTS)
OpenJDK 64-Bit Server VM Corretto-21.0.12.9.1 (build 21.0.12.1+9-LTS, mixed mode, sharing)
```

Install java

```
[root@ip-10-0-15-183 ~]# git --version
git version 2.50.1
```

Install git

```
[root@ip-10-0-15-183 ~]# useradd -m -s /bin/bash jenkins
[root@ip-10-0-15-183 ~]# mkdir -p /home/jenkins/.ssh
chown -R jenkins:jenkins /home/jenkins/.ssh
chmod 700 /home/jenkins/.ssh
[root@ip-10-0-15-183 ~]# passwd jenkins
Changing password for user jenkins.
New password:
Retype new password:
passwd: all authentication tokens updated successfully.
[root@ip-10-0-15-183 ~]# id jenkins
uid=1001(jenkins) gid=1001(jenkins) groups=1001(jenkins)
[root@ip-10-0-15-183 ~]# systemctl enable --now sshd
[root@ip-10-0-15-183 ~]# hostname -I
10.0.15.183
[root@ip-10-0-15-183 ~]# sudo -u jenkins mkdir -p /var/lib/jenkins/.ssh
mkdir: cannot create directory ‘/var/lib/jenkins’: Permission denied
[root@ip-10-0-15-183 ~]# sudo mkdir -p /var/lib/jenkins/.ssh
sudo chown -R jenkins:jenkins /var/lib/jenkins/.ssh
sudo chmod 700 /var/lib/jenkins/.ssh
```

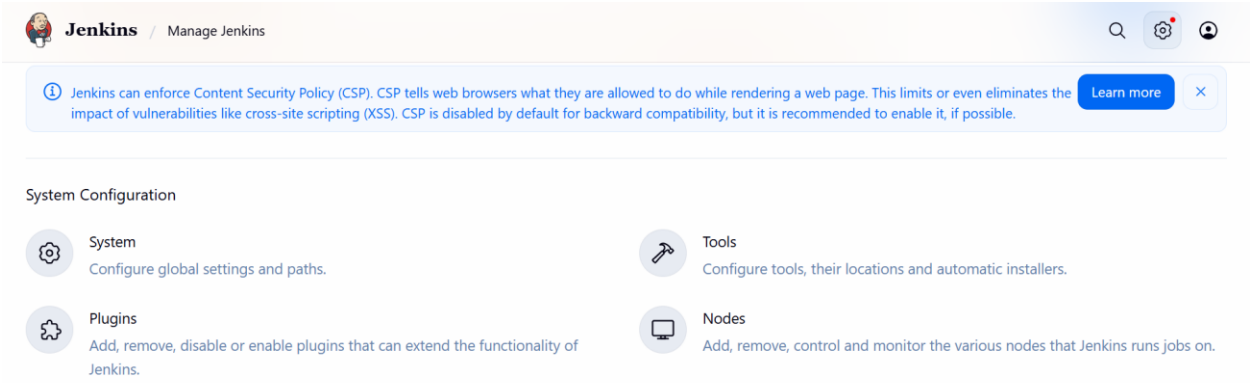
and in the jenkins-EC2, run

```
[root@ip-10-0-6-42 hiring-pipeline]# sudo -u jenkins ssh-keygen -t rsa -b 4096
Generating public/private rsa key pair.
Enter file in which to save the key (/var/lib/jenkins/.ssh/id_rsa):
Enter passphrase for "/var/lib/jenkins/.ssh/id_rsa" (empty for no passphrase):
Enter same passphrase again:
Your identification has been saved in /var/lib/jenkins/.ssh/id_rsa
Your public key has been saved in /var/lib/jenkins/.ssh/id_rsa.pub
The key fingerprint is:
SHA256:mlwW7cZCA0Zw6473NG3hIorPD7047HRdjIzm6eXmD2k jenkins@ip-10-0-6-42.ec2.internal
The key's randomart image is:
+---[RSA 4096]----+
|    .o=          |
|     o o .       |
|      . + .      |
|     . + B       |
|      + S B      |
|     B B B .     |
|   .+ % E +      |
|   +oB O.=       |
|  .o*o+o+..      |
+----[SHA256]-----+
```

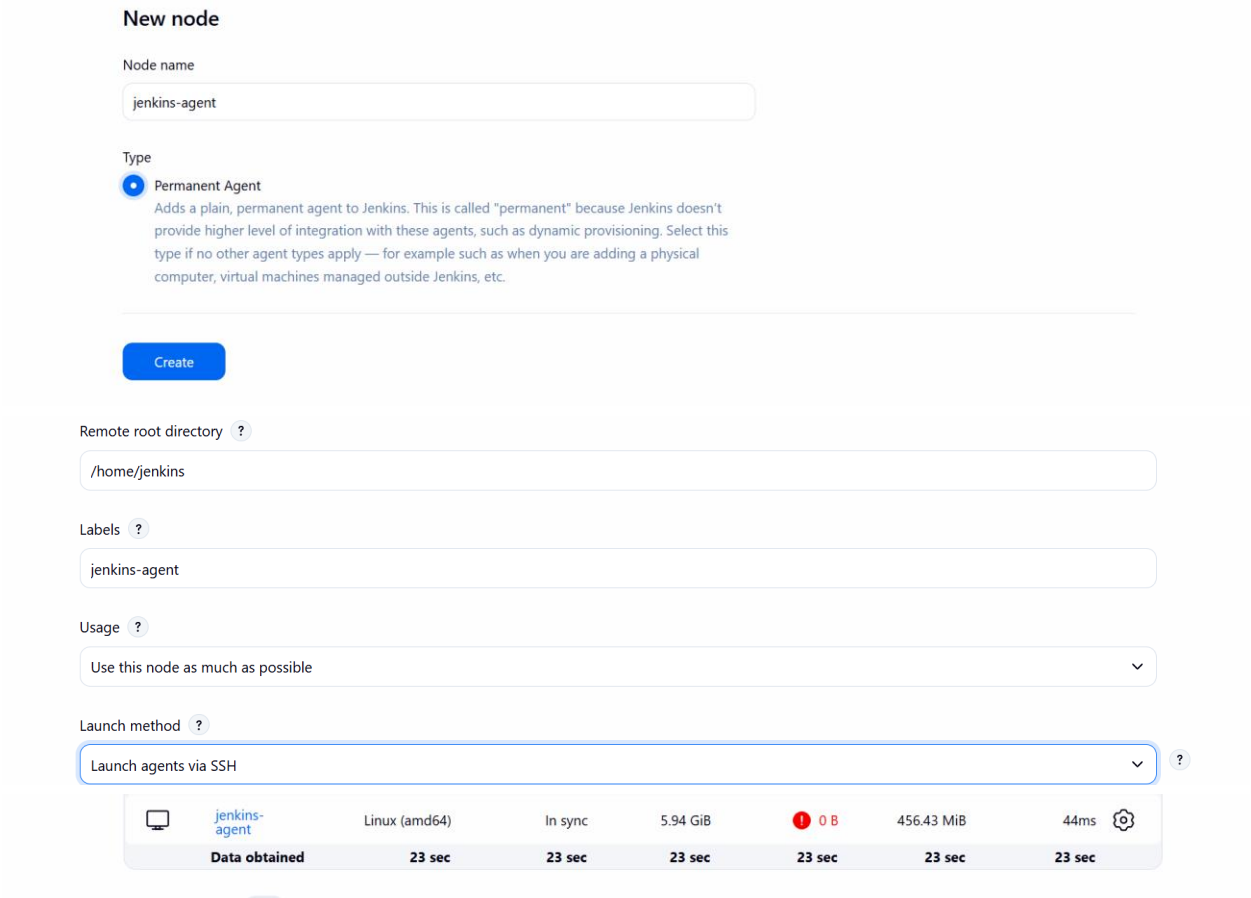
```
[root@ip-10-0-15-183 ~]# chown -R jenkins:jenkins /home/jenkins/.ssh
chmod 700 /home/jenkins/.ssh
chmod 600 /home/jenkins/.ssh/authorized_keys
```

```
[root@ip-10-0-6-42 hiring-pipeline]# sudo -u jenkins ssh -o StrictHostKeyChecking=no jenkins@10.0.15.183 'hostname'
Warning: Permanently added '10.0.15.183' (ED25519) to the list of known hosts.
ip-10-0-15-183.ec2.internal
[root@ip-10-0-6-42 hiring-pipeline]#
```

So the ssh connection is working



Choose nodes



The jenkins is connected and online

In the script change from

pipeline {

agent any

To

agent {

label 'jenkins-agent'

}

```
+ mvn clean package -DskipTests
/home/jenkins/workspace/hiring-pipeline@tmp/durable-66d3abaa/script.sh.copy: line 1: mvn
[Pipeline] }
[Pipeline] // stage
[Pipeline] }
[Pipeline] // node
[Pipeline] End of Pipeline
ERROR: script returned exit code 127
Finished: FAILURE
```

Install maven in the jenkins-agent

```
Complete!
[root@ip-10-0-15-183 ~]# mvn --version
Apache Maven 3.8.4 (Red Hat 3.8.4-3.amzn2023.0.5)
Maven home: /usr/share/maven
Java version: 17.0.20.1, vendor: Amazon.com Inc., runtime: /usr/lib/jvm/java-17-amazon-corretto.x86_64
Default locale: en, platform encoding: UTF-8
OS name: "linux", version: "6.18.51-120.163.amzn2023.x86_64", arch: "amd64", family: "unix"
```

